# When Should an LLM Issue Triager Stay Silent? Calibration, Drift, Rewording and Evidence in Issue Triage

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## Abstract

Issue-labeling services now apply LLM labels to open-source issues automatically. We evaluate one as a deployed, selective service: Qwen2.5-7B, zero-shot and with retrieved examples, against classical classifiers given the same input, on the NLBSE'24 benchmark and on issues the same five repositories received in 2025–26, after the model's training cutoff. TF-IDF is the most accurate on the benchmark (75.5%) but falls to 66.3% on 2025–26 issues and 60.4% on a new repository; the LLM is stable (70.9%, 75.3%, 68.0%). A confidence threshold tuned for 10% error among auto-labeled issues overshoots in every condition (up to 13.3% in distribution, 27.6% under shift). With thresholds fitted and certified on separate data, Learn-then-Test certifies 10% error only for TF-IDF (11% of issues auto-labeled); the LLM is first certified at a 20% target, and no certificate carries over to a new repository. Leave-one-repository-out calibration restores conformal coverage but not threshold risk. Controlled rewordings show that the phrase "sorry if this is a silly question" pushes the few-shot LLM toward the label *question* (+14.8 points vs. a neutral sentence), while newcomer status alone has a small effect; deferral keeps such flips from being auto-applied but halves the share of these reporters' issues labeled automatically. Evidence that the LLM's quotes drive its decisions is inconclusive. Input novelty detects a new repository within 13 requests with under 1% false alarms; the model's own confidence rarely notices. A record/replay regression suite catches injected calibration and retrieval bugs but barely catches prompt and preprocessing bugs.

## Keywords

issue triage, large language models, selective prediction, calibration, robustness, explanations, ML in production

## 1 Introduction

Open-source projects receive more issues than maintainers can read, and automatic labeling is one of the most requested forms of help. GitHub now ships an LLM-based labeler as an Action [8], and research has moved from fine-tuned classifiers to LLMs [3, 5]. Benchmarks report how often a labeler is right on a fixed snapshot [14]. A deployed labeler faces different questions. Which labels should it apply on its own, and which should it leave for a person? Does its promised error rate hold a year later, or on a repository it has never seen? Does it treat a report differently because the reporter writes in non-native English or apologizes for being new? And when it explains a decision, is the explanation what actually drove it?

We study these questions with a triage service built around Qwen2.5-7B, running locally, on the NLBSE'24 benchmark (3,000 issues from five large repositories, 2016–2023) and on issues the same repositories received in 2025–2026, after the model's training cutoff. We compare the LLM, zero-shot and with retrieved examples, against classical classifiers trained on the repositories' history, and ask:

- **RQ1** How do classical and LLM triagers compare in distribution, a year later, and on a new repository?
- **RQ2** Can calibrated deferral keep the error among auto-applied labels at a target, and does that promise survive drift?
- **RQ3** Do meaning-preserving changes in how an issue is written (no template, a newcomer sentence, non-native English) change the label, and which part of a newcomer sentence matters?
- **RQ4** Are the evidence quotes the LLM gives faithful to its decision?
- **RQ5** Which logged signal notices that the stream of issues has changed?

The answers are mixed in useful ways. A TF-IDF classifier trained on the history is the most accurate in distribution (75.5%) but drops to 66.3% on 2025–26 issues and 60.4% on a new repository; the LLM is less accurate in distribution (70.9%) but stable (75.3% and 68.0%). A confidence threshold tuned to 10% error on calibration data overshoots in every condition. When thresholds are fitted and certified on separate data with Learn-then-Test [2], a 10% target is certified only for TF-IDF, at 11% coverage; the few-shot LLM is first certified at 20% (63% of issues auto-labeled), and nothing transfers to a new repository. A sentence in which the reporter calls their post "a silly question" moves the few-shot LLM toward *question* by 14.8 points relative to a neutral sentence, while saying they are new moves it by 2.9 points. Deferral keeps almost all such flips from being auto-applied, but halves the share of these reporters' issues that are labeled automatically. Whether the LLM's evidence quotes drive its decisions remains inconclusive. A simple novelty signal, distance to the closest labeled history issue, detects a new repository within 13 requests; the model's own confidence rarely notices.

*Contributions.* (1) An evaluation of LLM issue triage as a deployed, selective service, including a post-cutoff 2025–26 set from the benchmark's own repositories; (2) evidence that common calibration recipes over-promise, a risk-control frontier showing what a few hundred calibration issues can certify, and a held-out-repository calibration protocol; (3) controlled rewording tests that separate the wording of a newcomer sentence from newcomer status, with their effect on maintainer workload; (4) a faithfulness test of LLM triage explanations; and (5) monitoring and record/replay regression tests for the service, all on one laptop.

## 2 Related Work

*Issue classification.* Ticket Tagger predicts GitHub issue types from text with fastText [11]. The NLBSE tool competitions turned the task into a shared benchmark; the 2024 edition provides 3,000 balanced issues from five repositories and a SetFit baseline [14]. Colavito et al. benchmark 22 decoder LLMs against fine-tuned encoders and find zero-shot LLMs less consistent across datasets [5]; Aracena et al. compare GPT-4o and DeepSeek-R1 on the same data [3]. GitHub now ships an LLM-based issue labeler as an Action [8]. These works report accuracy on a fixed benchmark; we ask how a deployed labeler should decide when *not* to label, and what happens after the benchmark period.

*Calibration and selective prediction.* Modern networks are overconfident, and temperature scaling is a simple fix [9]. Selective classification abstains below a confidence threshold chosen for a target risk [7]; split conformal prediction gives distribution-free coverage under exchangeability [1]. We test whether these guarantees survive the shifts a triage service meets in production.

*Behavioral testing, fairness and explanations.* CheckList tests NLP models with meaning-preserving perturbations [17]. LLMs answer users with lower English proficiency less accurately [15], which motivates testing whether a triager treats non-native reporters differently. ERASER measures whether rationales are faithful by deleting them [6]; we apply the same idea to the evidence quotes an LLM triager gives.

*ML in production.* Monitoring, calibration under drift, and regression tests are part of production readiness [4, 12].

## 3 Study Design

### 3.1 The triage service

The service receives a new GitHub issue and either applies one label (*bug*, *feature*, *question*) or leaves the issue for a maintainer. Bodies are preprocessed as a deployed service would: HTML comments left by issue templates are removed and the body is cut to 1,500 characters. We compare four classifiers, all given the same input (title and cleaned body). Two are classical: TF-IDF features with logistic regression, and `bge-small-en-v1.5` [18] embeddings of the same text with logistic regression. Two use Qwen2.5-7B-Instruct [16] (4-bit, MLX [10], one laptop): *zero-shot* with label definitions, and *few-shot*, which adds the six most similar labeled issues from the same repository, retrieved by embedding similarity. For the LLMs we read the next-token distribution after the prompt and take the probability mass on each label word, renormalized over the three labels; this costs one prefill pass and needs no verbalized confidence.

### 3.2 Data and conditions

*NLBSE'24* [14] supplies 3,000 issues (react, tensorflow, vscode, bitcoin, opencv; 2016–2023; 100 per repository and label in each half). We split its training half into a labeled history of 1,005 issues (training data for the classical models and retrieval pool for the few-shot LLM) and a calibration split of 495, and use its full test half (1,500 issues). We evaluate three conditions:

**In-distribution:** the NLBSE'24 test set.

**Fresh 2025–26:** issues created between January 2025 and October 2026 in the same five repositories, collected with the GitHub API and mapped to the three labels through each repository's own label names (1,430 single-type issues; a balanced sample of up to 40 per repository and label, 514 issues). History and calibration stay in 2016–2023, as for a service deployed and not retrained; the period also lies after the model's training cutoff, which removes benchmark leakage.

**New repository:** 750 test issues triaged as if their repository had just been onboarded: the few-shot model retrieves examples only from the other four repositories and the classical models are trained without the target repository. We calibrate this condition two ways: by *reusing* the in-distribution calibration (what a service does if it onboards a repository without recalibrating), and by *leave-one-repository-out* (LORO) calibration on the calibration issues of the other four repositories, each scored by the same cross-repository pipeline.

### 3.3 Selective automation

We split each classifier's calibration issues, balanced within repository and label, into a *fit* part (one third, 165 issues) and a *certification* part (two thirds, 330). The fit part sets the temperature [9] and a grid of candidate thresholds (its confidence quantiles at coverage levels 10%, 12.5%, …, 100%); the certification part is used only to choose or test thresholds. We compare five policies: *auto-all*; a *naive threshold*, the lowest confidence at which the auto-labeled certification issues have at most 10% errors [7]; a *risk-controlled threshold* from Learn-then-Test [2], which tests the pre-specified grid in order (most confident first) with an exact binomial test of $H_0$: risk > target at $\delta$=0.05 and stops at the first failure; *conformal*, which auto-labels only when the split-conformal prediction set at $\alpha$=0.1 (quantile from the certification part) has one label [1]; and *agreement*, which auto-labels only when the few-shot LLM and the embedding classifier agree. We also report the risk-controlled threshold for targets of 15%, 20% and 25%.

### 3.4 Rewording tests

From the test set we draw 210 probe issues (14 per repository and label) and create variants: *no template* (template headings and checklists removed), *non-native* (the prose rewritten as by a non-native English speaker; code blocks are masked during rewriting and restored), a *newcomer sentence* prepended to the body (*"Hi everyone, I'm new to this project and not very experienced, so sorry if this is a silly question or the wrong place to ask."*), and four controls that separate its parts: newcomer status only, an apology only (*"sorry if this is the wrong place to post this"*), the words *"sorry if this is a silly question"* only, and a neutral sentence of similar length. *No code* (fenced blocks and stack-trace lines removed) and *non-native + no code* remove information that may decide the label; we report them as reduced-input conditions, not as invariance tests. The non-native rewrite is produced by Llama-3.2-3B [13], a different model family from the classifier under test, and kept only if every code block is restored, the length stays within 0.5–1.8×, and at least 80% of technical tokens survive (161 of 210 after up to four attempts). For

**Table 1: Accuracy (%) by condition and cross-repository F1. Tokens: LLM prompt tokens per issue.**

| Approach | NLBSE'24 | 2025–26 | New repo | Tokens |
|---|---|---|---|---|
| TF-IDF + LR | 75.5 | 66.3 | 60.4 | – |
| Embedding + LR | 69.3 | 68.5 | 57.5 | – |
| LLM zero-shot | 68.5 | 71.2 | 68.4 | 0.4k |
| LLM few-shot | 70.9 | 75.3 | 68.0 | 1.1k |
| *Cross-repository F1 (NLBSE ranking metric)* | | | | |
| TF-IDF + LR | 0.754 | 0.583 | 0.592 | |
| Embedding + LR | 0.691 | 0.630 | 0.561 | |
| LLM zero-shot | 0.663 | 0.650 | 0.661 | |
| LLM few-shot | 0.694 | 0.682 | 0.657 | |

the meaning-preserving variants a label flip is a robustness failure; human validation of label preservation is pending (Section 6).

### 3.5 Explanations

For the probe issues the few-shot model is also asked for its label and two exact quotes from the issue as evidence. We check whether the quotes occur verbatim and run a deletion test [6]: the issue is re-scored, with the same retrieved examples, after deleting the quoted text and, as a control, after deleting random spans of the same total length (two seeds).

### 3.6 Statistics

Intervals are percentile bootstraps over issues (2,000 resamples). Monitoring follows the protocol of Section 4.5.

## 4 Results

### 4.1 RQ1: Accuracy in and out of distribution

Table 1. In distribution, TF-IDF with logistic regression, trained on 1,005 labeled issues, is the most accurate approach (75.5%); the few-shot LLM reaches 70.9% [68.6, 73.3] and zero-shot 68.5%. Out of distribution the order reverses. On 2025–26 issues TF-IDF drops to 66.3%, while the few-shot LLM reaches 75.3%. On a new repository TF-IDF and the embedding classifier fall to 60.4% and 57.5%; the LLMs lose at most 3 points (68.0%, 68.4%). The classical models learn repository- and period-specific vocabulary; the LLM relies on general knowledge of what bugs, features and questions look like.

Because the 2025–26 issues postdate the model's training data, they also act as a leakage control: the LLM is at least as accurate on them as on the 2016–2023 benchmark, so we see no sign that benchmark contamination inflates its results. The two samples are not directly comparable, though: label usage itself drifted. React, for example, labeled 200 bugs but only 9 feature requests and 5 questions in 2025–26, as it now routes questions elsewhere.

### 4.2 RQ2: Does calibrated deferral keep its promise?

Raw LLM label probabilities are almost always 0 or 1: temperature scaling needs $T$=7.9 (few-shot) and $T$=10.3 (zero-shot) and lowers expected calibration error on the certification issues from 0.23–0.30 to 0.05–0.07. The classical models are underconfident ($T \approx 0.7$).

*Naive thresholds over-promise.* Table 2 and Figure 1. A threshold set where the certification issues show at most 10% errors gives 10.2–13.3% errors on the test set, and under shift up to 20.1% (TF-IDF, 2025–26) and 27.6% (TF-IDF, new repository). The lowest feasible threshold sits where finite-sample noise happens to look favorable.

*Rigorous risk control certifies almost nothing at 10%.* With 330 certification issues, Learn-then-Test certifies a 10% threshold only for TF-IDF, which then auto-labels 11.3% of test issues with 6 errors in 170 (3.5%) and 6.8% of 2025–26 issues with 2 errors in 35. For the embedding classifier and both LLMs no threshold passes the first test, so every issue goes to a maintainer. Table 3 shows the trade-off. The few-shot LLM is first certified at a 20% target, where it auto-labels 62.9% of test issues at 19.1% actual risk and 69.8% of 2025–26 issues at 17.3%; the zero-shot LLM needs 25%. TF-IDF jumps from 17.4% coverage at a 15% target to 79.9% at 20%. The guarantee does not transfer to a new repository: with the reused calibration the few-shot LLM's 20% threshold gives 24.3% errors there, and LORO calibration on the other repositories certifies nothing below 25%. For a new repository the most our data support is that rigorous risk control is unaffordable at this calibration size; at a 10% target the reused TF-IDF threshold auto-labels 3 of 750 issues, too few to judge.

*Recalibration helps conformal sets, not thresholds.* Conformal prediction sets cover the true label in 86.9–88.0% of test cases (target 90%). On a new repository with reused calibration, coverage falls to 75.6% (TF-IDF) and 83.5% (embedding); LORO calibration restores it to 87.9% and 92.0% (LLMs: 88.1–88.3%). LORO calibration does not rescue the naive threshold: the few-shot LLM then auto-labels 19.2% of new-repository issues at 20.8% risk, TF-IDF 10.0% at 24.0%. An agreement gate between the LLM and the embedding classifier labels about two thirds of issues at 17.6–26.9% risk.

### 4.3 RQ3: Does the way an issue is written change its label?

Table 4 and Figure 2. TF-IDF is the most fragile to style and structure: removing template scaffolding or rewriting in non-native English flips 12.5–13.2% of its labels. The LLMs are more robust to these, but non-native English still costs the zero-shot LLM 6.9 points of accuracy and the few-shot LLM 4.4. Removing code is not meaning-preserving in the same sense, since a stack trace can be the decisive evidence for *bug*; we report it as reduced input. Without code and in non-native English, the zero-shot LLM loses 10.1 points.

*The newcomer sentence acts through the word "question".* The full newcomer sentence flips 16.2% of the few-shot LLM's labels, 14.8% toward *question*. The controls in Table 5 locate the effect. Relative to a neutral sentence, paired by issue, the phrase *"sorry if this is a silly question"* alone moves 14.8 points [10.0, 20.0] of issues to *question*, newcomer status alone 2.9 points [1.0, 5.2], and an apology alone nothing. The zero-shot LLM shows the same pattern at a smaller scale (6.7 points for the phrase). TF-IDF flips 1.4–7.1% under every added sentence, the neutral one included, and is not pushed toward *question*. The LLM reads the reporter's description of their own post as evidence about the issue's type; newcomer status itself has a small but measurable effect.

**Table 2: Selective automation: coverage / risk (%), i.e., share auto-labeled / errors among them; for the risk-controlled policy also (errors/auto-labeled). Target risk: 10%. "Reused": the in-distribution calibration applied to the new repository; "LORO": leave-one-repository-out calibration. Conformal: auto-label singleton sets.**

| Approach | Policy | NLBSE'24 | 2025–26 | New repo (reused) | New repo (LORO) |
|---|---|---|---|---|---|
| TF-IDF + LR | auto-all | 100.0 / 24.5 | 100.0 / 33.7 | 100.0 / 39.6 | 100.0 / 39.6 |
| | naive threshold | 60.1 / 12.5 | 53.1 / 20.1 | 39.2 / 27.6 | 10.0 / 24.0 |
| | risk-controlled | 11.3 / 3.5 (6/170) | 6.8 / 5.7 (2/35) | 0.4 / 33.3 (1/3) | 0.0 / – |
| | conformal | 68.1 / 15.7 | 67.9 / 23.5 | 58.3 / 32.0 | 25.1 / 23.9 |
| Embedding + LR | auto-all | 100.0 / 30.7 | 100.0 / 31.5 | 100.0 / 42.5 | 100.0 / 42.5 |
| | naive threshold | 31.9 / 10.2 | 27.8 / 11.9 | 20.5 / 22.1 | 0.0 / – |
| | risk-controlled | 0.0 / – | 0.0 / – | 0.0 / – | 0.0 / – |
| | conformal | 48.9 / 16.3 | 44.2 / 18.1 | 38.1 / 28.7 | 17.1 / 19.5 |
| LLM zero-shot | auto-all | 100.0 / 31.5 | 100.0 / 28.8 | 100.0 / 31.6 | 100.0 / 31.6 |
| | naive threshold | 12.4 / 12.4 | 23.7 / 8.2 | 10.5 / 13.9 | 13.1 / 15.3 |
| | risk-controlled | 0.0 / – | 0.0 / – | 0.0 / – | 0.0 / – |
| | conformal | 33.1 / 19.1 | 43.8 / 12.4 | 30.5 / 18.8 | 31.5 / 19.9 |
| LLM few-shot | auto-all | 100.0 / 29.1 | 100.0 / 24.7 | 100.0 / 32.0 | 100.0 / 32.0 |
| | naive threshold | 36.6 / 13.3 | 44.6 / 10.0 | 32.0 / 19.2 | 19.2 / 20.8 |
| | risk-controlled | 0.0 / – | 0.0 / – | 0.0 / – | 0.0 / – |
| | conformal | 57.2 / 18.5 | 63.6 / 17.1 | 55.3 / 23.9 | 38.5 / 20.1 |
| LLM few-shot + Emb. | agreement | 67.4 / 18.2 | 72.8 / 17.6 | 62.0 / 26.9 | – |

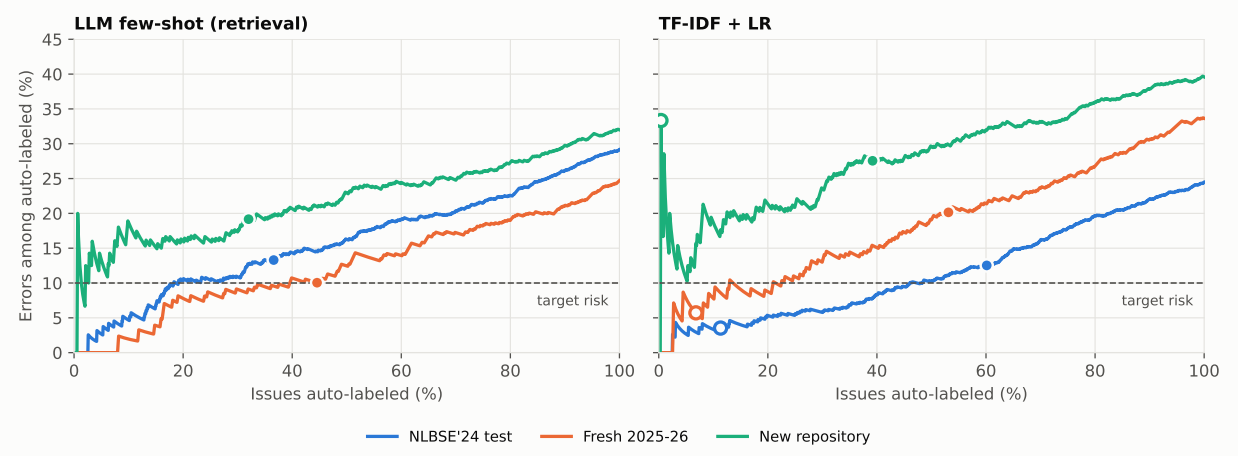



**Figure 1: Risk–coverage curves: error among the most confident issues as more are auto-labeled. Filled dots: naive threshold; open dots: risk-controlled threshold (none certified for the LLM at 10%). Dashed line: target.**

*Deferral moves the cost to maintainers.* Under the naive threshold, the share of variants where both versions would be auto-labeled with *different* labels is 0–1.9% for every classifier: flips happen among issues the service is already unsure about. The price is review work. The full newcomer sentence halves the share of the few-shot LLM's auto-labeled issues (33.8% to 16.2%), the "silly question" phrase alone cuts it to 13.3%, and non-native rewriting lowers it from 35.8% to 32.1%. Reporters who write this way are labeled automatically less often and wait for a human more often.

## 4.4 RQ4: Are the explanations faithful?

Asked for a label and two exact quotes as evidence, the few-shot LLM quoted the issue in 178 of 210 cases; 93.1% of its 331 quotes occur verbatim, and the explained label matches the scored label in 92.9% of issues. On 164 issues with locatable quotes (on average 130 of 921 body characters), deleting the quoted evidence lowered the calibrated probability of the predicted label by 0.024, against 0.014 for deleting random text of the same length (Wilcoxon $p$=0.115), and changed the label in 6.1% of issues against 3.0%. The evidence is inconclusive: the quotes are real, but we cannot show that they matter more to the decision than other text of the same length.

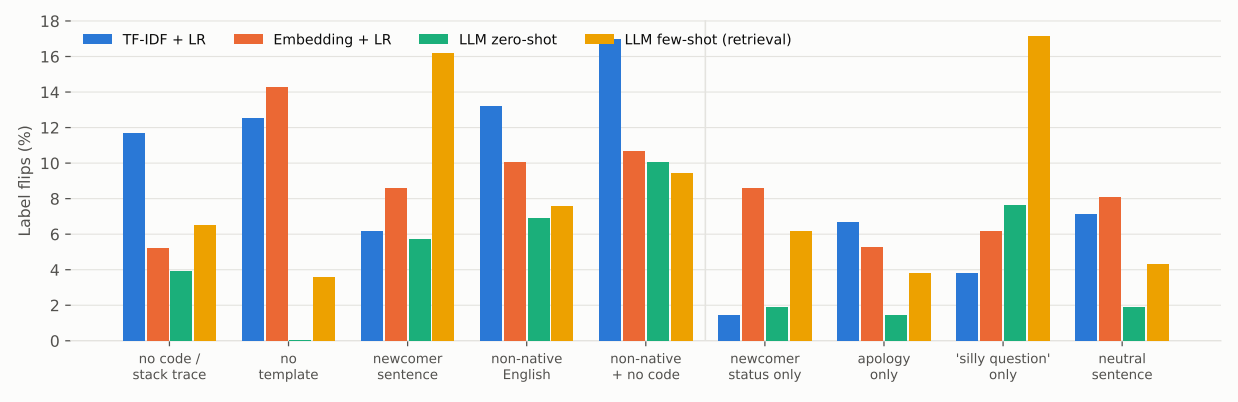



**Figure 2: Label flips under rewording. Right of the divider: parts of the newcomer sentence and a neutral control.**

**Table 3: Risk-controlled threshold certified for each target ($\delta$=0.05): coverage / actual risk (%). 0.0 / –: no threshold certified.**

| Approach | Target | NLBSE'24 | 2025–26 | New (reused) | New (LORO) |
|---|---|---|---|---|---|
| TF-IDF + LR | 10% | 11.3 / 3.5 | 6.8 / 5.7 | 0.4 / 33.3 | 0.0 / – |
| | 15% | 17.4 / 4.2 | 14.4 / 9.5 | 1.7 / 15.4 | 0.0 / – |
| | 20% | 79.9 / 19.6 | 69.6 / 23.5 | 62.3 / 32.3 | 2.1 / 12.5 |
| | 25% | 99.9 / 24.4 | 100.0 / 33.7 | 99.6 / 39.6 | 17.6 / 27.3 |
| Embedding + LR | 10% | 0.0 / – | 0.0 / – | 0.0 / – | 0.0 / – |
| | 15% | 0.0 / – | 0.0 / – | 0.0 / – | 0.0 / – |
| | 20% | 38.1 / 12.9 | 33.9 / 16.1 | 27.9 / 23.0 | 0.0 / – |
| | 25% | 68.4 / 22.4 | 63.8 / 24.1 | 56.1 / 33.5 | 3.1 / 21.7 |
| LLM zero-shot | 10% | 0.0 / – | 0.0 / – | 0.0 / – | 0.0 / – |
| | 15% | 0.0 / – | 0.0 / – | 0.0 / – | 0.0 / – |
| | 20% | 0.0 / – | 0.0 / – | 0.0 / – | 2.8 / 4.8 |
| | 25% | 58.9 / 23.2 | 71.6 / 20.1 | 57.6 / 23.6 | 22.7 / 26.5 |
| LLM few-shot | 10% | 0.0 / – | 0.0 / – | 0.0 / – | 0.0 / – |
| | 15% | 0.0 / – | 0.0 / – | 0.0 / – | 0.0 / – |
| | 20% | 62.9 / 19.1 | 69.8 / 17.3 | 60.3 / 24.3 | 0.0 / – |
| | 25% | 78.0 / 22.4 | 82.1 / 19.7 | 78.0 / 26.7 | 11.6 / 24.1 |

**Table 4: Rewording of 210 probe issues (*n*: applicable variants): label flips, accuracy change, and share auto-labeled by the naive threshold. †Reduced-input conditions that may remove decisive evidence.**

| | No code† | No templ. | Newcomer | Non-native | NN + no code† |
|---|---|---|---|---|---|
| *n* | 77 | 112 | 210 | 159 | 159 |
| *Label flips (%)* | | | | | |
| TF-IDF + LR | 11.7 | 12.5 | 6.2 | 13.2 | 17.0 |
| Embedding + LR | 5.2 | 14.3 | 8.6 | 10.1 | 10.7 |
| LLM zero-shot | 3.9 | 0.0 | 5.7 | 6.9 | 10.1 |
| LLM few-shot | 6.5 | 3.6 | 16.2 | 7.5 | 9.4 |
| *Accuracy change, original → variant (points)* | | | | | |
| TF-IDF + LR | -2.6 | +0.9 | -2.9 | -4.4 | -6.9 |
| Embedding + LR | -5.2 | +0.0 | +2.9 | -2.5 | -4.4 |
| LLM zero-shot | -3.9 | +0.0 | -2.4 | -6.9 | -10.1 |
| LLM few-shot | +0.0 | -1.8 | -3.3 | -4.4 | -5.0 |
| *Auto-labeled (%), original → variant* | | | | | |
| TF-IDF + LR | 64→61 | 71→65 | 60→55 | 63→60 | 63→62 |
| Embedding + LR | 32→43 | 38→39 | 36→38 | 33→33 | 33→39 |
| LLM zero-shot | 17→16 | 16→13 | 10→4 | 11→12 | 11→9 |
| LLM few-shot | 36→30 | 45→42 | 34→16 | 36→32 | 36→30 |

## 4.5 RQ5: Which signal notices a changed stream?

We simulate 300 streams per shift: 400 in-distribution requests, then 200 from 2025–26, from a repository without history, or in non-native English. A signal alarms when its rate over the last 50 requests exceeds the mean plus three standard deviations of a separate clean calibration stream, drawn from a disjoint half of the test issues; false alarms are counted on the evaluation stream before the shift. Table 6 and Figure 3: *novelty*, one minus the cosine similarity to the closest labeled history issue the service may retrieve from, detects a new repository a median of 13 requests after the shift (all runs) and 2025–26 traffic after 44 (96% of runs), with under 1% false alarms. The model's own signals — share handed to a human, low confidence — detect these shifts in at most 27% of runs. LLM–classifier disagreement detects a new repository in 63% of runs but raises 5–7% false alarms. No signal reliably notices the non-native stream.

## 4.6 Regression tests

A 40-case suite (issues auto-labeled correctly, wrong predictions that must be handed off, and non-native rewrites that must keep their label) replays the full pipeline from recorded model scores in seconds. Injecting plausible bugs and replaying live: removing temperature scaling fails 13 cases (must-hand-off issues that are then auto-labeled), retrieving examples from other repositories fails 8, and dropping retrieval fails 6. Removing the label definitions from the prompt (2 cases) and cutting the body to 300 characters (1 case) are barely caught; the suite needs cases aimed at preprocessing and prompt content.

## 5 Lessons for Practitioners

*Benchmark accuracy is the wrong selection criterion.* The most accurate classifier on the benchmark was the least accurate a year later and on a new repository. Evaluate a labeler on issues from after its training and calibration data, and on a held-out repository, before choosing it.

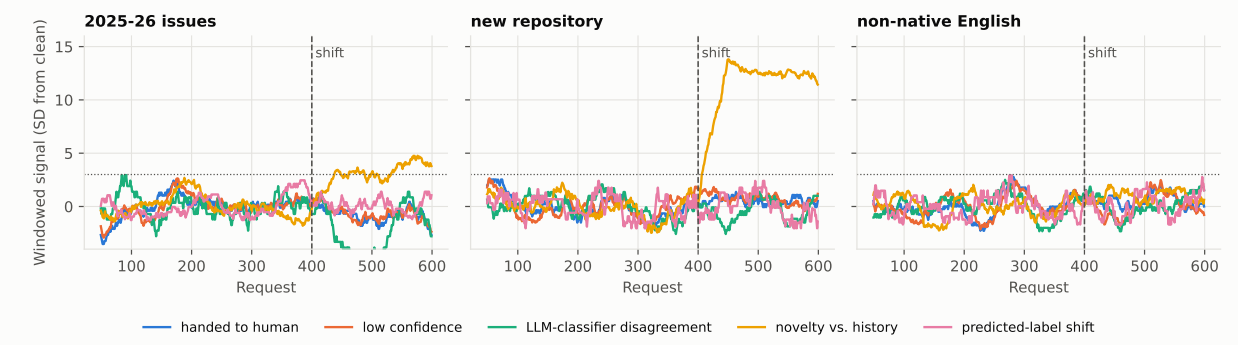



**Figure 3: Windowed signals in one simulated stream per shift, standardized against the clean segment for display; the dotted line is three standard deviations.**

**Table 5: Which part of the newcomer sentence matters? Flips, flips to *question*, the paired difference in flips to *question* relative to the neutral sentence (percentage points, 95% bootstrap CI), and share auto-labeled (%).**

| Approach | Sentence | Flips | →question | Δ vs. neutral | Auto-lab. |
|---|---|---|---|---|---|
| LLM zero-shot | Full newcomer sentence | 5.7 | 4.8 | +4.8 [+1.9, +7.6] | 4.3 |
| | Newcomer status only | 1.9 | 0.5 | +0.5 [+0.0, +1.4] | 10.0 |
| | Apology only | 1.4 | 0.0 | +0.0 [+0.0, +0.0] | 9.5 |
| | "Silly question" only | 7.6 | 6.7 | +6.7 [+3.3, +10.5] | 3.8 |
| | Neutral sentence | 1.9 | 0.0 | – | 10.5 |
| LLM few-shot | Full newcomer sentence | 16.2 | 14.8 | +13.3 [+9.0, +18.6] | 16.2 |
| | Newcomer status only | 6.2 | 4.3 | +2.9 [+1.0, +5.2] | 29.5 |
| | Apology only | 3.8 | 1.4 | +0.0 [-1.9, +2.4] | 31.0 |
| | "Silly question" only | 17.1 | 16.2 | +14.8 [+10.0, +20.0] | 13.3 |
| | Neutral sentence | 4.3 | 1.4 | – | 32.4 |
| TF-IDF + LR | Full newcomer sentence | 6.2 | 2.4 | -4.3 [-7.1, -1.9] | 55.2 |
| | Newcomer status only | 1.4 | 0.0 | -6.7 [-10.0, -3.8] | 57.6 |
| | Apology only | 6.7 | 1.0 | -5.7 [-9.5, -2.9] | 59.5 |
| | "Silly question" only | 3.8 | 3.3 | -3.3 [-5.7, -1.0] | 59.0 |
| | Neutral sentence | 7.1 | 6.7 | – | 59.0 |

**Table 6: Detection delay after a shift: median requests to the first alarm (share of 300 runs detected); –: detected in fewer than half. Last row: highest false-alarm rate across shifts.**

| Shift | Handoff | Low conf. | Disagree | Novelty | Label shift |
|---|---|---|---|---|---|
| 2025–26 issues | – | – | – | 44 (96%) | – |
| New repository | – | – | 64 (63%) | 13 (100%) | – |
| Non-native | – | – | – | – | – |
| *False alarms* | 0.5% | 1.0% | 7.0% | 0.7% | 1.7% |

*A calibrated threshold is not an error guarantee.* Tuning the threshold to the target on a few hundred calibration issues over-promised in every condition. A risk-controlling procedure needs its threshold candidates fixed before certification, and with 330 certification issues it certified a 10% target only for the classical classifier and at low coverage. Choose the target from a frontier like Table 3, or budget for far more labeled calibration issues. A certificate holds only for data exchangeable with the calibration set: recalibrate, on that repository's own labeled issues, when a repository is onboarded.

*Calibration across repositories fixes sets, not thresholds.* Leave-one-repository-out calibration brought conformal coverage back to about 88–92% on a new repository, but naive thresholds still overshot. Prediction sets, which keep a human in the loop for ambiguous issues, transfer better than a single auto-label threshold.

*Deferral protects labels, not reporters.* Flips under rewording concentrate where the service is unsure, so almost none are auto-applied. But a reporter who calls their post "a silly question" is auto-labeled half as often and waits for a human instead. Measure workload and wait time by reporter group, not only flips, and keep reporter self-descriptions out of the classifier's input where possible.

*Show evidence as context, not as cause.* The LLM's quotes are genuine, but we could not show that they matter more than other text; a triage UI should present them as relevant passages, not as the reason for the label.

*Monitor inputs, not only the model.* A nearest-neighbor novelty score over the labeled history caught a new repository within 13 requests at under 1% false alarms; the model's confidence did not, and disagreement between models alarmed falsely in 5–7% of clean windows.

## 6 Threats to Validity

*Construct.* Labels are the repositories' own type labels, which mix issue content with project policy (React routes questions elsewhere). The 2025–26 set maps each repository's label names to the three types; issues with several type labels are excluded. Rewrites are synthetic. The non-native rewrite was validated automatically (code blocks restored, length and technical tokens preserved; 161 of 210 valid) and occasionally alters template fields; a human validation of meaning preservation on a sample (`report/audit/rewrite_audit.csv`) is pending. Removing code is treated as reduced input, not as a meaning-preserving change. The cue sentences are one wording each; other phrasings of the same status may behave differently.

*Internal.* The calibration issues come from the benchmark's training half and the test set from its test half, as in the competition. Temperature, threshold candidates and naive thresholds are fitted

on one third of the calibration issues and certified on the other two thirds; nothing is tuned on test data. All classifiers see the same 1,500 characters of each issue. Probabilities are read from the first output token; a different prompt or label wording would change them.

*External.* One LLM, five large repositories, three labels. Smaller projects, other label sets, and stronger models (including fine-tuned encoders such as the SetFit competition baseline) may behave differently. The monitoring results come from simulated streams mixed from our pools, not from a live deployment. All runs used one laptop.

*Conclusion validity.* Rewording tests use 77–210 variants per kind; differences of a few points between classifiers are within noise, while the "silly question" effect on the few-shot LLM is not (paired bootstrap CI [10.0, 20.0]). New-repository results for risk-controlled thresholds rest on very few auto-labeled issues (TF-IDF: 3 of 750 at the 10% target) and support no conclusion beyond the low coverage. The explanation test is underpowered ($p$=0.115).

## 7 Conclusion

Treating issue triage as a deployed, selective service changes what matters. The LLM was less accurate than a simple classifier on the benchmark but held up a year later and on a new repository. With an honest protocol, a few hundred calibration issues certify a 10% error rate for almost no issues, and no certificate carried over to a new repository. A reporter's description of their own post, not their newcomer status, moved the LLM toward *question*; deferral kept those flips from being applied but shifted the work to maintainers. The cheapest monitor, input novelty, was the most useful. The harness, data collection scripts, recorded model scores and regression suite are available for replication.

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